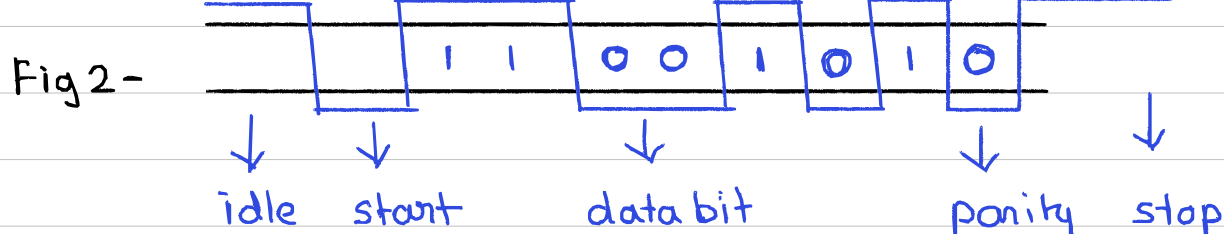
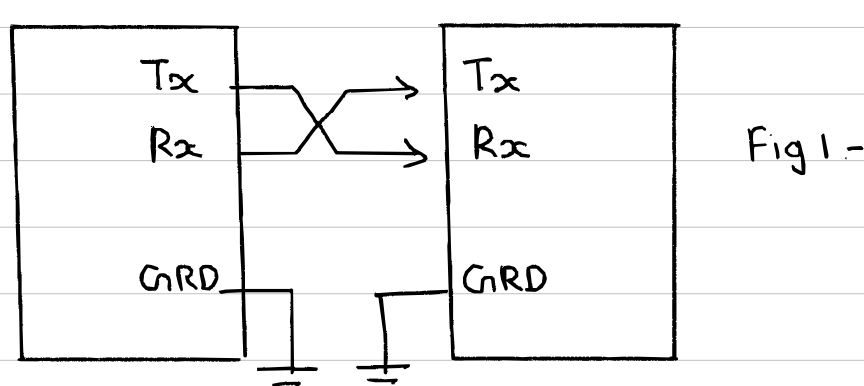


2B: Elucidate SPI, I2C and UART protocol and its application?

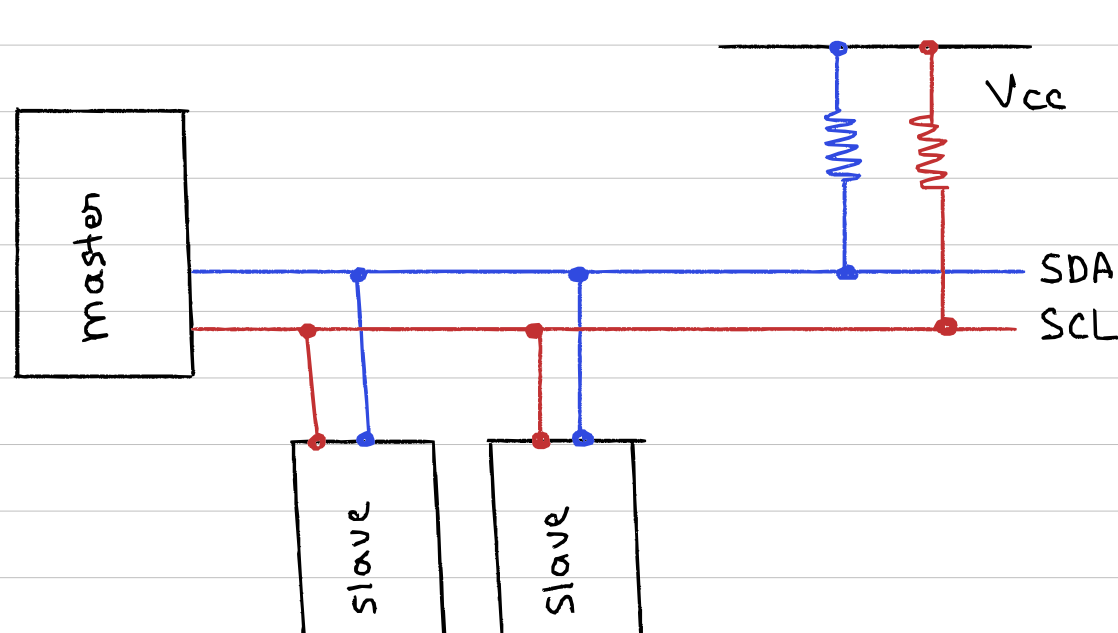
UART- Universal Asynchronous Receiver and Transmitter

- protocol for exchanging serial data between 2 devices
- Used only 2 wires
 1. Tx → Transmitter
 2. Rx → Receiver
- can be simplex, half duplex and full duplex
- UART is asynchronous. The Tx and Rx do not share a common clock so therefore transmit at the same known speed.
- Popularity of UART however decreasing but still it is used for lower speed low throughput application
- UART frame consist of :-
 1. start/stop bit
 2. Data bit
 3. Parity bit (Optional)



I2C- Inter-Integrated circuit

- Developed in 1982 by Philips (now NXP)
- primarily used for short distance data communication
- Synchronous master - slave protocol
Both master and slave can send / receive data
Bidirectional, half duplex
can run at different speed
- Only 2 wires
 1. SCL - Serial clock
 2. SDA - Serial Data
- I2C frame consist of :-
 1. start / stop
 2. Slave address
 3. Read and write Acknowledgement
 4. Data bit
 5. Ack

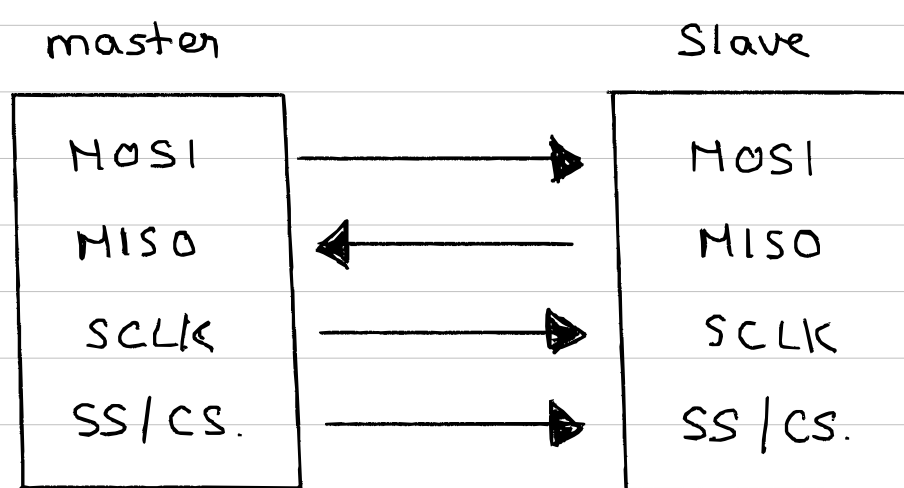


Application:

1. microcontroller to peripheral communication
2. Sensor interfacing in embedded system
3. communication inside consumer electronic
4. Automotive Application
5. multi master system

SPI- Serial peripheral Interface

- Common Synchronous communication protocol used by many device. For example SD card and RFID card reader.
- Sending data without interruption
- master / Slave relationship, where master control data transfer
- Design to transfer data btw various IC chip at very high speed, hence short bus connection is used to maintain stability
- Data and control Line of SPI are :-
 1. MOSI - master out slave in
 2. MISO - master in slave out
 3. Serial clock (SCLK)
 4. Slave Select (SS)



Application

1. High speed Sensor and converter
→ Accelerometer
2. Display module
→ TFT colour displays
3. Memory devices (Flash and SD cards)